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A Growing Visible Light Leak in the 94 Å Channel of the Atmospheric Imaging Assembly

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ABSTRACT

The 94 Å channel of the Atmospheric Imaging Assembly (AIA) on the Solar Dynamics Observatory (SDO) is the instrument's main view of coronal plasma hotter than a few megakelvin, and with the other extreme ultraviolet (EUV) channels it underlies most differential emission measure (DEM) analyses of the corona. We find that the channel also records visible light from the photosphere. The leak is a sharp, limb-darkened image of the solar disk with a broad halo, and since each visible photon frees a single electron, it adds almost none of the photon noise that an EUV signal of the same size would. It is absent from the 304 Å channel, which shares the telescope and the detector, it is attenuated more than the EUV by the thicker of the two focal-plane filters, and it can be seen directly as SDO passes into the shadow of the Earth, where it sets **XX** after the EUV has been absorbed, flattened by refraction like a sunset. In **XX** such eclipses between **XX** and **XX**, the leak at disk center grew from **XX** to **XX**, exponentially since **XX** with a doubling time of **XX**, and it is now **XX** of the quiet-Sun signal in the channel. Cross-calibration with the Solar Ultraviolet Imager (SUVI) on GOES-18 and GOES-19 and the noise of the channel itself give the same history. Left in the data, the leak appears in DEMs as hot plasma in the quiet Sun and in coronal holes, it biases maps of Fe XVIII, and it hides part of the degradation of the channel from its calibration. We give a model of the leak and its history from which it can be subtracted.

Keywords: Solar instruments (1499), Calibration (2179), Solar extreme ultraviolet emission (1493), Solar corona (1483), Quiet sun (1322), Solar coronal holes (1484), Eclipses (442), Limb darkening (922)

1. INTRODUCTION

Outline.

- **The channel.** The Atmospheric Imaging Assembly (AIA) (J. R. Lemen et al. 2012) images the corona in seven extreme ultraviolet (EUV) channels from the Solar Dynamics Observatory (SDO) (W. D. Pesnell et al. 2012). The 94 Å channel sees Fe XVIII from plasma near 7 MK, and cooler lines of Fe IX and Fe X which atomic databases underestimate (P. Testa et al. 2012), so its temperature response was corrected empirically against the EUV Variability Experiment (EVE) (T. N. Woods et al. 2012; P. F. Boerner et al. 2014).
- **What depends on it.** It is the main constraint on hot plasma in differential emission measure (DEM) inversions (I. G. Hannah & E. P. Kontar 2012; J. Plowman et al. 2013; M. C. M. Cheung et al. 2015; H. Morgan & J. Pickering 2019; J. Plowman & A. Caspi 2020), and maps of Fe XVIII are made by subtracting the cool part of it predicted from the other channels (H. P. Warren et al. 2012; G. Del Zanna 2013).
- **Calibration over the mission.** The degradation of each channel is tracked against EVE and its successors (P. Boerner et al. 2012; P. F. Boerner et al. 2014; L. F. G. Dos Santos et al. 2021) and distributed in `aiapy` (W. Barnes et al. 2020). A signal that grows with time reads as a channel which degrades more slowly than it does.
- **A precedent.** The zirconium filters of the 94 Å channels transmit a little visible light (S. K. Tadikonda et al. 2019), and pinholes in the entrance filters of the Solar Ultraviolet Imager (SUVI) on GOES-17 leak visible light, most in its 94 Å channel (J. M. Darnel et al. 2022). Check that Darnel2022 describes the GOES-17 leak; otherwise cite the SUVI L1b readme.
- **Symptoms already in the literature.** In coronal holes the 94 Å signal matches what remains behind an occulter, and was taken for stray light (J. Saqri et al. 2020; S. G. Heinemann et al. 2021). A survey of quiet-Sun DEMs ended in 2018 because the reconstructions after it were poor (S. Purkhart & A. M. Veronig 2022), and some inversions drop the channel (S. J. Hofmeister et al. 2025). Coronal holes lose their contrast in it (S. M. Lezzi et al. 2023), and it is brighter than a DEM constrained in X-rays allows (P. S. Athiray & A. R. Winebarger 2024).

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- **Roadmap.** Section 2 finds a noiseless signal in the channel, Section 3 measures it directly in eclipses over the mission, Section 4 confirms it with the focal-plane filters, the other channels, and SUVI, Section 5 follows it into thermal diagnostics and calibration, Section 6 gives a correction, and Section 7 concludes.

2. PHOTON NOISE

Outline.

- **Data.** Runs of consecutive level 1 images at the full cadence of the channel, returned to the units of the detector by multiplying by the flat field they were divided by.
- **Method.** The variance of the second difference of three consecutive frames, which cancels the linear evolution of the Sun, against the signal estimated from frames independent of it. The read noise comes from the corners of the charge-coupled device (CCD) beyond the field stop.
- **The noiseless signal.** Off the limb the variance is the read noise plus a constant times the signal, as expected. On the disk it is offset: part of the signal carries no photon noise. Its map is a limb-darkened disk which vanishes off the limb and does not follow the structure of the corona in any channel.
- **Its history.** The noiseless signal at disk center, from runs spread over the mission.
- **Figures.** Variance against signal on and off the disk; the map of the noiseless signal; its history.

3. EARTH ECLIPSES

Outline.

- **Eclipse seasons.** SDO passes behind the Earth once a day for a few weeks twice a year, and the seasons drift earlier from year to year.
- **Ingress.** The atmosphere absorbs the EUV high above the surface, while the visible disk sets only near it, so for a while each part of the disk is seen in visible light alone. The 171 Å channel marks where the EUV has gone.
- **Method.** Each bin of the disk on a clock started by its own 171 Å edge; a forward model of a limb-darkened disk setting behind the atmosphere, seen through a point spread function (PSF) with a sharp core and broad wings, plus whatever EUV is left, fit to the frames in which the EUV is gone. The frames in the umbra are the dark.
- **Results.** The leak at disk center in **XX** eclipses from **XX** to **XX**: from **XX** to **XX**, with a doubling time of **XX** since **XX**. Its limb darkening, the sharpness of its limb, and the halo beyond it.
- **Caveats.** The core and the wings of the PSF are degenerate, though the light they deliver to the disk is not; one ingress fell during a flare; before **XX** there is only a faint diffuse glow.
- **Figures.** An ingress in 94 Å and 171 Å; the leak at disk center against time; its radial profile.

4. INDEPENDENT CHECKS

Outline.

- **The focal-plane filters.** On the one day since 2024 that the channel observed through its thick focal-plane filter, the leak fell by more than the EUV did, so it passes through that filter and is light rather than an artifact of the detector or its calibration.
- **The other channels.** The 304 Å channel shares the telescope and the CCD but has no leak, so the leak is in the filters of the 94 Å half of the aperture. What the noise says, and cannot say, about the 131, 171, 193, 211, and 335 Å channels.
- **SUVI.** A joint fit of AIA to SUVI on GOES-16, GOES-18, and GOES-19 with a limb-darkened term on the disk and a constant gives the same history, the constant being the halo.

5. CONSEQUENCES

Outline.

- **Differential emission measures.** The fraction of the DEM of coronal holes and the quiet Sun above 3 MK, early in the mission and now, with and without the leak subtracted. Subtracting it restores the quiet Sun, but coronal holes stay hotter than they were early in the mission.
- **Fe XVIII.** In quiet regions the leak survives the subtraction of the cool part of the channel.
- **Calibration.** In full-disk comparisons a growing leak reads as rising sensitivity. The correction tables now hold the sensitivity of the channel constant.
- **Published results.** Which of the results of Section 1 the leak accounts for, and how much.

6. A CORRECTION

Outline.

- **Model.** A limb-darkened disk seen through the PSF of the leak, scaled by an amplitude interpolated between eclipses, in the units of the detector and divided by the flat field.
- **Uncertainty.** From the scatter about the exponential and between the methods.
- **Keeping it current.** Two eclipse seasons a year measure the leak as it grows. Decide where the correction lives: `aiapy`, `sdo`, or this package.

7. CONCLUSIONS

Outline.

- The 94 Å channel records a growing image of the photosphere in visible light.
- Four independent measurements agree on its size, shape, and history.
- What it means for hot plasma outside active regions since about 2018, and what to do about it.
- Where the leak enters the instrument, and how to tell, for the AIA team.

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